

# Rodrigo Pimenta

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## EXPERIENCE

### Systems Engineer III

Nov 2021 – Present

*Hughes Network Systems*

*Germantown, MD*

- Second-highest committer of ~15 engineers on the satellite gateway automation platform: 400+ commits across 53 delivered tickets spanning monitoring, alerting, self-healing, and reporting for five ground-segment platforms
- Root-caused and eliminated a platform-wide logging defect that silently discarded log records, cutting the log footprint from 19.68 GB to 58 MB and reclaiming ~25 GB in production; built audit tooling that swept all 146 scheduled jobs for the same pattern instead of fixing them individually
- Root-caused two production failures: a 738 MB core dump traced with gdb to an unguarded deserialization path in a vendor gateway component, and a 177-process leak traced to a library deadlock at interpreter teardown; delivered both as actionable defect reports and shipped the supervision module that closed the leak
- Designed the platform's shared logging entry point, now used by every script and device module; introduced continuous integration to the organization; migrated the action-audit data model with 26,857 rows reconciled exactly
- Containerized Python automation services with Docker for deployment across Linux hosts and to a Kubernetes cluster, backed by a Patroni-managed high-availability PostgreSQL pair
- Accelerated deployment of 20 data center racks from four weeks to two using Ansible, and automated L2/L3 network configuration to reduce configuration errors by 90% across 1,000+ satellite sites
- Lead analyst on infrastructure failure investigations and 24/7 on-call against a 99.9% availability SLA; rollback owner in formal change control, building incident timelines from core-dump metadata and process logs

### Software Engineering Fellow

Jan 2021 – Nov 2021

*Pathrise*

*Remote*

- Fellowship in algorithms, data structures, and technical interviewing — weekly Python problem sets and mentor-led mock interviews across trees, graphs, heaps, recursion, and dynamic programming

### Mass Communication Specialist, E-3 (Active Duty)

Jan 2020 – Jan 2021

*United States Navy Reserve*

*Great Lakes, IL / Fort Meade, MD*

- Completed basic training and the Defense Information School (DINFOS) at Fort Meade, MD; honorably discharged

### Technology Analyst

Jun 2018 – Dec 2019

*Bank of America*

*Charlotte, NC*

- Developed PowerShell and Python scripts to automate software validation and hardware testing across 20+ ATM models in the enterprise lab environment
- Built a reusable CI/CD deployment framework in Jenkins that standardized release processes and moved builds from weekly to hourly, reducing environment setup time by 15%

## SYSTEMS BUILT

### Ceph Storage Self-Healing Monitor | *Python, uv, Fabric/SSH, PostgreSQL, Grafana, pytest*

- Sole author (77 of 79 commits, 6,586 lines). Monitors 24 storage nodes across 8 clusters on a 3-minute cycle. Automated metadata-server failover gated by a four-verdict standby-capacity assessment so it never fails over into a worse node, with outcomes verified against the cluster log to distinguish a real failover from a same-node reclaim. Append-only event history across 7 PostgreSQL tables with additive-only schema migration, a 19-panel Grafana dashboard, and a pytest suite in CI

### Config Integrity Self-Heal | *Python, asyncio, SFTP, PostgreSQL*

- Sole author (30 of 30 commits, 3,594 lines). Detects configuration drift and corruption across the gateway fleet over SFTP and restarts the affected service, modernizing a blocking-I/O workload onto asyncio with semaphore-bounded fan-out rather than process pools

### Orbital Watch | *Python, Cloudflare Workers, CI/CD, pytest* | [orbital.rodrigopimenta.com](https://orbital.rodrigopimenta.com)

- Deployed satellite pass-prediction and space-weather dashboard built on the thesis that a system should report its own degradation: each upstream source is isolated, falls back cache → committed snapshot, and is classified fresh/stale/failed on the page. 142 tests, offline-deterministic build, scheduled rebuild on a Cloudflare Worker cron with a dead-man check

## EDUCATION

### University of Maryland, A. James Clark School of Engineering

College Park, MD

*Bachelor of Science in Computer Engineering | College Park Scholars*

*Aug 2014 – Dec 2018*

## TECHNICAL SKILLS

**Languages:** Python, SQL, Bash, PowerShell, JavaScript

**Automation, CI/CD & Testing:** Ansible, Molecule, Jenkins, Bitbucket Pipelines, Git, pytest, uv, ruff, pre-commit, Fabric

**Cloud, Data & Observability:** Google Cloud (BigQuery), PostgreSQL (Patroni HA), Grafana, pandas, SQLAlchemy, psycopg2

**Systems:** Linux/RHEL, Docker, Kubernetes, Ceph, asyncio, gdb / core-dump analysis, VMware, 24/7 on-call | **AI tooling:** Kiro, Cursor, Claude Code